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WATER ELECTROLYSIS SYSTEM

Abstract

A water electrolysis system includes: a water electrolysis device for electrolyzing water; a gas-liquid separator for performing gas-liquid separation of a mixed fluid of hydrogen gas and water, the mixed fluid being led out from the water electrolysis device; a dehumidifier for dehumidifying the hydrogen gas separated from the mixed fluid by the gas-liquid separator; a delivery path for delivering the hydrogen gas dehumidified by the dehumidifier; a humidifier for humidifying the hydrogen gas delivered through the delivery path; and a compression device for compressing the hydrogen gas humidified by the humidifier.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-026106 filed on Feb. 26, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a water electrolysis system.

Description of the Related Art

[0003] In recent years, technological development has been conducted on power systems that contribute to energy efficiency in order to ensure that more people have access to affordable, reliable, sustainable and modern energy.

[0004] JP 2022-083098 A discloses a water electrolysis system including a water electrolysis device, a gas-liquid separator, a delivery path, and a compression device. The water electrolysis device electrolyzes water. The gas-liquid separator separates a mixed fluid of hydrogen gas and liquid water, which is led out from the water electrolysis device, into gas and liquid. The delivery path is configured to deliver the hydrogen gas separated from the mixed fluid by the gas-liquid separator. The compression device compresses the hydrogen gas guided from the delivery path.

SUMMARY OF THE INVENTION

[0005] There has been a demand for a more satisfactory water electrolysis system.

[0006] The present invention has the object of solving the aforementioned problem.

[0007] According to an aspect of the present disclosure, there is provided a water electrolysis system comprising: a water electrolysis device configured to electrolyze water; a gas-liquid separator configured to perform gas-liquid separation of a mixed fluid of hydrogen gas and water, the mixed fluid being led out from the water electrolysis device; a dehumidifier configured to dehumidify the hydrogen gas separated from the mixed fluid by the gas-liquid separator; a delivery path configured to deliver the hydrogen gas dehumidified by the dehumidifier; a humidifier configured to humidify the hydrogen gas delivered through the delivery path; and a compression device configured to compress the hydrogen gas humidified by the humidifier.

[0008] According to the present disclosure, a more satisfactory water electrolysis system can be provided.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic configuration diagram of a water electrolysis system according to an embodiment of the present disclosure; and

[0011] FIG. 2 is a schematic configuration diagram of the water electrolysis system including a water supply unit according to a modification.

DETAILED DESCRIPTION OF THE INVENTION

[0012] In a water electrolysis system, hydrogen gas separated from a mixed fluid by a gas-liquid separator contains water vapor. In this case, since hydrogen gas introduced into a compression device through a delivery path also contains water vapor, the electrolyte membranes constituting

the compression device can be humidified by the water vapor. However, in a case where the temperature around the delivery path is relatively low, for example, condensed water may be generated in the delivery path, and the humidity of the hydrogen gas introduced into the compression device may not be appropriately controlled. Further, the delivery path may be blocked by the condensed water. The present disclosure has been made in view of such a problem, and can provide a water electrolysis system capable of suppressing the delivery path from being blocked by condensed water and also capable of appropriately controlling the humidity of hydrogen gas introduced into the compression device.

[0013] FIG. 1 is a schematic configuration diagram of a water electrolysis system **10** according to an embodiment of the present disclosure. The water electrolysis system **10** can form part of a circulative renewable energy system **12**, for example. The circulative renewable energy system **12** is a system in which the water electrolysis system **10** that generates oxygen gas and hydrogen gas by electrolyzing water, and a fuel cell system that generates electricity and water by an electrochemical reaction between oxygen gas and hydrogen gas, are combined. In the circulative renewable energy system **12**, the water electrolysis system **10** generates, by using water generated in the fuel cell system, oxygen gas and hydrogen gas required for power generation of the fuel cell system.

[0014] Such a circulative renewable energy system **12** can be installed, for example, on the ground or on the moon's surface. Further, the circulative renewable energy system **12** can also be mounted on a satellite such as an International Space Station (ISS). The water electrolysis system **10** is not limited to the example of being provided in the circulative renewable energy system **12**, and may be provided in a hydrogen station, for example.

[0015] As shown in FIG. 1, the water electrolysis system **10** includes a water electrolysis device **14**, a gas-liquid separator **16**, a dehumidifier **18**, a first hydrogen supply path **20**, a delivery path **22**, a humidifier **24**, a second hydrogen supply path **26**, a compression device (electrochemical compression device) **28**, and a water supply unit **30**.

[0016] The water electrolysis device **14** generates oxygen gas and hydrogen gas by electrolyzing water (pure water). The water electrolysis device **14** is, for example, a solid polymer water electrolysis device. The water electrolysis device **14** may be an alkaline water electrolysis device, a solid oxide water electrolysis device, or the like.

[0017] The water electrolysis device **14** includes a water electrolysis stack **32** and a water electrolysis power supply **34**. The water electrolysis stack **32** includes a plurality of water electrolysis cells **36** stacked on each other. The water electrolysis power supply **34** is a DC power supply. The water electrolysis power supply **34** applies a voltage between an anode current collector and a cathode current collector of each of the water electrolysis cells **36**.

[0018] The detailed illustration of the water electrolysis cells **36** will be omitted. In each of the water electrolysis cells **36**, water is supplied to the cathode flow field of the water electrolysis cell **36**, and the water electrolysis power supply **34** applies a voltage between the anode current collector and the cathode current collector, thereby electrolyzing water. As a result, oxygen gas is generated in the anode flow field of the water electrolysis cell **36**, and hydrogen gas is generated in the cathode flow field of the water electrolysis cell **36**. An ion exchange membrane of the water electrolysis cell **36** prevents the oxygen gas generated in the anode flow field from passing through to the cathode flow field. Consequently, the oxygen gas can be stored in the anode flow field, and therefore, high-pressure oxygen gas can be generated in the anode flow field.

[0019] The high-pressure oxygen gas generated in the anode flow field of each of the water electrolysis cells **36** is led out to the outside of the water electrolysis device **14** through an oxygen gas lead-out path **38**. The oxygen gas lead-out path **38** guides the oxygen gas to, for example, an oxygen gas tank of the fuel cell system (not shown). The hydrogen gas generated in the cathode flow field is guided to the gas-liquid separator **16** through a hydrogen gas lead-out path **40** together with water (liquid water).

[0020] The gas-liquid separator **16** separates a mixed fluid of the hydrogen gas and the water, which is guided through the hydrogen gas lead-out path **40**, into gas and liquid. The gas-liquid separator **16** includes a reservoir portion **42** for storing water (liquid water) separated from the mixed fluid. Water (liquid water) is supplied to the gas-liquid separator **16** from a water introduction path **44**. For example, water generated by power generation of the fuel cell system (not shown) is guided to the water introduction path **44**. It should be noted that water produced by a pure water production apparatus (not shown) may be guided to the water introduction path **44**. The water stored in the reservoir portion **42** is supplied to the water electrolysis device **14** via the water supply unit **30**. The detailed configuration of the water supply unit **30** will be described later. The hydrogen gas separated from the mixed fluid by the gas-liquid separator **16** contains water vapor. [0021] The first hydrogen supply path **20** connects the gas-liquid separator **16** and the dehumidifier **18**. The first hydrogen supply path **20** guides, to the dehumidifier **18**, the hydrogen gas separated from the mixed fluid by the gas-liquid separator **16**. A deoxygenation catalyst **46** is provided in the first hydrogen supply path **20**. The deoxygenation catalyst **46** removes oxygen from the hydrogen gas. The deoxygenation catalyst **46** contains, for example, platinum.

[0022] The dehumidifier **18** dehumidifies the hydrogen gas guided from the first hydrogen supply path **20**. In other words, the dehumidifier **18** dehumidifies the hydrogen gas separated from the mixed fluid by the gas-liquid separator **16**. The dehumidifier **18** is a membrane dehumidifier. The membrane dehumidifier includes a plurality of hollow fiber membranes **48**, and an outer tube **50** for accommodating these hollow fiber membranes **48**.

[0023] Each of the hollow fiber membranes **48** is made of a material that allows water vapor to pass therethrough but does not allow hydrogen gas to pass therethrough. The hydrogen gas (containing water vapor) guided from the first hydrogen supply path **20** flows through the hollow fiber membranes **48**. A dehumidifying space **52** for circulating dry hydrogen gas is formed outside the hollow fiber membranes **48**. The dehumidifying space **52** is covered with the outer tube **50**. In this case, the water vapor pressure inside the hollow fiber membranes **48** becomes higher than the water vapor pressure in the dehumidifying space **52**, and therefore, due to the water vapor pressure difference, the water vapor contained in the hydrogen gas permeates through the hollow fiber membranes **48** and moves to the dehumidifying space **52**. Consequently, the hydrogen gas is dehumidified. In such a membrane dehumidifier, the temperature change of the hydrogen gas can be suppressed as compared to a case where the hydrogen gas is dehumidified by being cooled.

[0024] The delivery path **22** is configured to deliver the hydrogen gas dehumidified by the dehumidifier **18** to the humidifier **24**. The delivery path **22** is a relatively long flow path. The delivery path **22** is longer than the first hydrogen supply path **20** and the second hydrogen supply path **26**. Further, the length of the delivery path **22** may be greater than the total length of the first hydrogen supply path **20** and the second hydrogen supply path **26**. The delivery path **22** is provided with a hydrogen pump **54** and a branch path **56**. The hydrogen pump **54** sends, toward the compression device **28** (the humidifier **24**), the hydrogen gas guided from the dehumidifier **18**. Therefore, it is possible to suppress the generation of condensed water in the hydrogen pump **54**. The branch path **56** branches off from a portion of the delivery path **22** that is located on the downstream side of the hydrogen pump **54**. Therefore, it is possible to prevent the hydrogen gas containing water vapor from flowing backward from the downstream side of the branch path **56** (the gas-liquid separator **16**).

[0025] The branch path **56** guides, to the dehumidifier **18**, the hydrogen gas flowing through the delivery path **22**. Specifically, the branch path **56** guides the dry hydrogen gas to the dehumidifying space **52** of the dehumidifier **18**. The hydrogen gas introduced from the branch path **56** into the dehumidifying space **52** is returned to the gas-liquid separator **16** through a return flow path **58** together with the water vapor (water) contained in the dehumidifying space **52**. As a result, the dehumidifying space **52** can be maintained in a dry state by the dry hydrogen gas. That is, since the water vapor pressure in the dehumidifying space **52** can be made lower than the water vapor

pressure inside the hollow fiber membranes **48**, the hydrogen gas can be efficiently dehumidified.

[0026] The branch path **56** is provided with an oxygen sensor **60** for measuring the oxygen concentration in the dry hydrogen gas flowing through the branch path **56**. In this case, it is possible to suppress the generation of condensed water in the oxygen sensor **60**.

[0027] The humidifier **24** humidifies the dry hydrogen gas delivered through the delivery path **22**. The humidifier **24** is a membrane humidifier. The membrane humidifier has a similar configuration to the membrane dehumidifier. That is, the membrane humidifier includes a plurality of hollow fiber membranes **62**, and an outer tube **64** for accommodating the hollow fiber membranes **62**.

[0028] Each of the hollow fiber membranes **62** is made of a material that allows water vapor to pass therethrough but does not allow hydrogen gas to pass therethrough. The dry hydrogen gas guided from the delivery path **22** flows through the hollow fiber membranes **62**. A humidifying space **66** for circulating water vapor is formed outside the hollow fiber membranes **62**. The humidifying space **66** is covered with the outer tube **64**. In this case, the water vapor pressure in the humidifying space **66** becomes higher than the water vapor pressure inside the hollow fiber membranes **62**, and therefore, due to the water vapor pressure difference, the water vapor in the humidifying space **66** permeates through the hollow fiber membranes **62** and moves into the hollow fiber membranes **62**. Consequently, the hydrogen gas is humidified.

[0029] The second hydrogen supply path **26** connects the humidifier **24** and the compression device **28**. The second hydrogen supply path **26** is configured to introduce the hydrogen gas humidified by the humidifier **24** into the compression device **28**.

[0030] The compression device **28** is an electrochemical hydrogen pump that electrochemically compresses the hydrogen gas introduced from the second hydrogen supply path **26**. The compression device **28** includes a compression stack **68** and a compression power supply **70**. The compression stack **68** includes a plurality of compression cells **72** stacked on each other. The compression power supply **70** is a DC power supply. The compression power supply **70** applies a voltage between an anode current collector and a cathode current collector of each of the compression cells **72**.

[0031] The detailed illustration of the compression cells **72** will be omitted. In each of the compression cells **72**, humidified hydrogen gas is supplied to the anode flow field, and the compression power supply **70** applies a voltage between the anode current collector and the cathode current collector. Consequently, hydrogen ions are generated in the anode flow field, and the hydrogen ions pass through an ion exchange membrane of the compression cell **72** and are guided to the cathode flow field. In the cathode flow field, hydrogen ions are combined to generate hydrogen gas. The ion exchange membrane of the compression cell **72** prevents the hydrogen gas generated in the cathode flow field from passing through to the anode flow field. Consequently, the hydrogen gas can be stored in the cathode flow field, and therefore, high-pressure hydrogen gas can be generated in the cathode flow field.

[0032] The high-pressure hydrogen gas generated in the cathode flow field of each of the compression cells **72** is led out to the outside of the compression device **28** through a hydrogen gas discharge path **74**. The hydrogen gas discharge path **74** guides the hydrogen gas to, for example, a hydrogen gas tank of the fuel cell system (not shown).

[0033] The water vapor contained in the hydrogen gas flowing through the anode flow field of each of the compression cells **72** humidifies the ion exchange membrane of the compression cell **72**. As a result, excessive drying of the ion exchange membrane of the compression cell **72** can be suppressed. Unreacted hydrogen gas guided from the anode flow field of each of the compression cells **72** is returned to the gas-liquid separator **16** through a circulation flow path **76** together with the water vapor.

[0034] The water supply unit **30** includes a first water supply path **78**, a second water supply path **80**, a water pump **82**, and a heat exchanger **84**. The first water supply path **78** connects the gas-liquid separator **16** and the humidifier **24**. The first water supply path **78** guides the water (liquid

water) stored in the reservoir portion **42** of the gas-liquid separator **16**, to the humidifier **24**. Specifically, the first water supply path **78** guides the water to the humidifying space **66** of the humidifier **24**. Water vapor can be generated in the humidifying space **66** by the water introduced from the first water supply path **78** into the humidifying space **66**. As a result, since the water vapor pressure in the humidifying space **66** can be made higher than the water vapor pressure inside the hollow fiber membranes **62**, the hydrogen gas can be efficiently humidified. In addition, a heating device (not shown) may be provided in the first water supply path **78**. Consequently, the temperature of the hydrogen gas in the humidifying space **66** can be increased while the hydrogen gas is humidified.

[0035] The second water supply path **80** connects the humidifier **24** and the water electrolysis device **14**. The second water supply path **80** guides, to the water electrolysis device **14**, the water (liquid water) that has flowed through the humidifier **24**. The water pump **82** is provided in the second water supply path **80**. The water pump **82** sends, to the water electrolysis device **14**, the water flowing through the second water supply path **80**. It should be noted that the location where the water pump **82** is provided can be set as appropriate, and may be, for example, the first water supply path **78**.

[0036] The heat exchanger **84** is provided in a portion of the second water supply path **80** that is located on the downstream side of the water pump **82**. It should be noted that the location where the heat exchanger **84** is provided can be set as appropriate, and for example, the heat exchanger **84** may be provided in a portion of the second water supply path **80** that is located on the upstream side of the water pump **82**, or may be provided in the first water supply path **78**. The heat exchanger **84** adjusts the temperature of water to be supplied to the water electrolysis device **14**. Consequently, water at an appropriate temperature is supplied to the water electrolysis device **14**.

[0037] The water stored in the reservoir portion **42** of the gas-liquid separator **16** is circulated by the water pump **82** through the first water supply path **78**, the humidifier **24**, the second water supply path **80**, the water electrolysis device **14**, and the hydrogen gas lead-out path **40**. Therefore, water whose temperature has been adjusted by the heat exchanger **84** is introduced into the humidifying space **66** of the humidifier **24**. Accordingly, by using water vapor generated from the water, the humidifier **24** can adjust the temperature of the hydrogen gas to an appropriate temperature.

[0038] Next, the operation of the water electrolysis system **10** will be described. In the water electrolysis system **10**, when the water pump **82** is driven, water (liquid water) stored in the reservoir portion **42** of the gas-liquid separator **16** is supplied to the water electrolysis device **14** through the first water supply path **78**, the humidifier **24**, and the second water supply path **80**. In the water electrolysis device **14**, high-pressure oxygen gas and low-pressure hydrogen gas are generated by electrolysis of the water. The high-pressure oxygen gas is guided to the outside of the water electrolysis device **14** through the oxygen gas lead-out path **38**. A mixed fluid of the low-pressure hydrogen gas and the water (liquid water) is guided to the gas-liquid separator **16** through the hydrogen gas lead-out path **40**.

[0039] The mixed fluid guided from the hydrogen gas lead-out path **40** to the gas-liquid separator **16** is separated into hydrogen gas (containing water vapor) and water (liquid water) by the gas-liquid separator **16**. The water separated from the mixed fluid is stored in the reservoir portion **42** and reused.

[0040] The hydrogen gas separated from the mixed fluid and containing water vapor is drawn into the dehumidifier **18** by the hydrogen pump **54** through the first hydrogen supply path **20** and the deoxygenation catalyst **46**. The dehumidifier **18** dehumidifies the hydrogen gas. The hydrogen gas dehumidified by the dehumidifier **18** (dry hydrogen gas) is sent to the humidifier **24** by the hydrogen pump **54** through the delivery path **22**. The hydrogen gas flowing through the delivery path **22** is dry, and therefore, even in a case where the water electrolysis system **10** is operated in a low-temperature environment, for example, it is possible to suppress the generation of condensed

water in the delivery path **22**. The humidifier **24** appropriately humidifies the hydrogen gas and adjusts the temperature of the hydrogen gas to an appropriate temperature. The hydrogen gas that has been humidified and subjected to the temperature adjustment by the humidifier **24** is introduced into the compression device **28** through the second hydrogen supply path **26**.

[0041] The compression device **28** compresses the humidified hydrogen gas. The compressed high-pressure hydrogen gas is guided to the outside of the compression device **28** through the hydrogen gas discharge path **74**. The water vapor humidifies the electrolyte membranes of the compression cells **72**. The hydrogen gas that has not reacted in the compression device **28** is guided to the gas-liquid separator **16** through the circulation flow path **76** together with the excess water vapor.

[0042] According to the present embodiment, the hydrogen gas separated from the mixed fluid by the gas-liquid separator **16** is dehumidified by the dehumidifier **18**, and the dehumidified hydrogen gas is guided to the delivery path **22**. Therefore, even when the temperature around the delivery path **22** is relatively low, it is possible to suppress the generation of condensed water inside the delivery path **22**. Further, the hydrogen gas delivered through the delivery path **22** is humidified by the humidifier **24**, and the humidified hydrogen gas is introduced into the compression device **28**. As a result, the humidity of the hydrogen gas introduced into the compression device **28** can be appropriately controlled. Therefore, a more satisfactory water electrolysis system **10** can be provided.

(Modification)

[0043] Next, a water supply unit **30a** according to a modification will be described. In the water supply unit **30a** according to the modification, the same components as those of the above-described water supply unit **30** are denoted by the same reference numerals, and the detailed description thereof will be omitted.

[0044] FIG. **2** is a schematic configuration diagram of the water electrolysis system **10** including the water supply unit **30a** according to the modification. As shown in FIG. **2**, the water supply unit **30a** includes a first portion **86** for supplying water to the humidifier **24**, and a second portion **88** for supplying water to the water electrolysis device **14**. The first portion **86** and the second portion **88** are separated from each other. The first portion **86** includes the first water supply path **78**, a water return flow path **90**, and a first water pump **92**. The water return flow path **90** connects the humidifier **24** and the gas-liquid separator **16**. The water return flow path **90** guides, to the gas-liquid separator **16**, the water (liquid water) that has flowed through the humidifier **24**. The first water pump **92** is provided in the water return flow path **90**. The first water pump **92** sends, to the gas-liquid separator **16**, the water flowing through the water return flow path **90**. It should be noted that the location where the first water pump **92** is provided can be set as appropriate, and may be the first water supply path **78**.

[0045] The second portion **88** includes a second water supply path **94**, a second water pump **96**, and the heat exchanger **84**. The second water supply path **94** connects the gas-liquid separator **16** and the water electrolysis device **14**. The second water supply path **94** guides the water stored in the reservoir portion **42** of the gas-liquid separator **16**, to the water electrolysis device **14**. The second water pump **96** is provided in the second water supply path **94**. The second water pump **96** sends, to the water electrolysis device **14**, the water flowing through the second water supply path **94**.

[0046] In the present modification, the water supply unit **30a** is divided into the first portion **86** for supplying water to the humidifier **24** and the second portion **88** for supplying water to the water electrolysis device **14**. Further, the first portion **86** includes the first water pump **92**, and the second portion **88** includes the second water pump **96**. Therefore, the flow rate of the water supplied to the humidifier **24** and the flow rate of the water supplied to the water electrolysis device **14** can be individually adjusted in accordance with the operating state of the water electrolysis system **10**, the ambient temperature, and the like.

[0047] The present embodiment is not limited to the configuration described above. The water

electrolysis system **10** may not include the deoxygenation catalyst **46**. The oxygen sensor **60** may be provided in the delivery path **22**.

[0048] The following supplementary notes are further disclosed in relation to the above-described embodiment.

Supplementary Note 1

[0049] The water electrolysis system (**10**) of the present disclosure includes: the water electrolysis device (**14**) configured to electrolyze water; the gas-liquid separator (**16**) configured to perform gas-liquid separation of a mixed fluid of hydrogen gas and water, the mixed fluid being led out from the water electrolysis device; the dehumidifier (**18**) configured to dehumidify the hydrogen gas separated from the mixed fluid by the gas-liquid separator; the delivery path (**22**) configured to deliver the hydrogen gas dehumidified by the dehumidifier; the humidifier (**24**) configured to humidify the hydrogen gas delivered through the delivery path; and the compression device (**28**) configured to compress the hydrogen gas humidified by the humidifier.

[0050] According to such a configuration, the hydrogen gas separated from the mixed fluid by the gas-liquid separator is dehumidified by the dehumidifier, and the dehumidified hydrogen gas is guided to the delivery path. Therefore, even when the temperature around the delivery path is relatively low, it is possible to suppress the generation of condensed water inside the delivery path. Further, the hydrogen gas delivered through the delivery path is humidified by the humidifier, and the humidified hydrogen gas is introduced into the compression device. As a result, the humidity of the hydrogen gas introduced into the compression device can be appropriately controlled.

Therefore, a more satisfactory water electrolysis system can be provided.

Supplementary Note 2

[0051] In the water electrolysis system according to Supplementary Note 1, the water obtained by dehumidifying the hydrogen gas using the dehumidifier may be guided to any of the gas-liquid separator or the humidifier.

[0052] According to such a configuration, the water obtained by dehumidifying the hydrogen gas using the dehumidifier can be reused.

Supplementary Note 3

[0053] In the water electrolysis system according to Supplementary Note 1 or 2, the dehumidifier may be a membrane dehumidifier.

[0054] According to such a configuration, it is possible to suppress a change in the temperature of the hydrogen gas when the hydrogen gas is dehumidified.

Supplementary Note 4

[0055] In the water electrolysis system according to any one of Supplementary Notes 1 to 3, water obtained by any of the gas-liquid separator or the dehumidifier may be guided to the humidifier.

[0056] According to such a configuration, water can be efficiently supplied to the humidifier.

Supplementary Note 5

[0057] In the water electrolysis system according to Supplementary Note 4, the gas-liquid separator may include the reservoir portion (**42**) configured to store water, and the water electrolysis system may further include the water supply unit (**30, 30a**) configured to supply the water stored in the reservoir portion to the humidifier and the water electrolysis device.

[0058] According to such a configuration, it is not necessary to separately provide a portion for storing water to be supplied to the humidifier and a portion for storing water to be supplied to the water electrolysis device, and therefore, the configuration of the water electrolysis system can be made compact.

Supplementary Note 6

[0059] In the water electrolysis system according to Supplementary Note 5, the water supply unit may include the heat exchanger (**84**) configured to adjust the temperature of the water to be supplied to the humidifier and the water electrolysis device.

[0060] According to such a configuration, the temperature of the water to be supplied to the water

electrolysis device can be adjusted by the heat exchanger. As a result, the humidifier can humidify the hydrogen gas guided from the delivery path and adjust the temperature of the hydrogen gas to an appropriate temperature.

Supplementary Note 7

[0061] In the water electrolysis system according to any one of Supplementary Notes 1 to 6, the humidifier may be a membrane humidifier.

[0062] According to such a configuration, the hydrogen gas can be efficiently humidified.

Supplementary Note 8

[0063] In the water electrolysis system according to any one of Supplementary Notes 1 to 7, the delivery path may be provided with the pump (54) configured to send, to the compression device, the hydrogen gas guided from the dehumidifier.

[0064] According to such a configuration, since the hydrogen gas dehumidified by the dehumidifier (dry hydrogen gas) flows through the pump, it is possible to suppress the generation of condensed water in the pump. As a result, it is possible to suppress a decrease in durability of the pump due to the condensed water.

Supplementary Note 9

[0065] The water electrolysis system according to any one of Supplementary Notes 1 to 8 may further include the oxygen sensor (60) configured to measure the oxygen concentration in the hydrogen gas flowing through the delivery path.

[0066] According to such a configuration, since the hydrogen gas dehumidified by the dehumidifier (dry hydrogen gas) flows through the oxygen sensor, it is possible to suppress the generation of condensed water in the oxygen sensor. As a result, it is possible to suppress a decrease in durability of the oxygen sensor due to the condensed water.

Supplementary Note 10

[0067] In the water electrolysis system according to Supplementary Note 8, the delivery path may be provided with the branch path (56) that branches off from a portion of the delivery path that is located on the downstream side of the pump, and the water obtained by the dehumidifier may be returned to the gas-liquid separator by the hydrogen gas flowing through the branch path.

[0068] According to such a configuration, the water generated in the dehumidifier can be returned to the gas-liquid separator by the hydrogen gas sent from the pump and guided to the branch path. As a result, since it is not necessary to provide a new pump for returning the water generated in the dehumidifier to the gas-liquid separator, the cost of the water electrolysis system can be reduced and the size thereof can be reduced.

[0069] Although the present disclosure has been described in detail, the present disclosure is not limited to the above-described individual embodiments. Various additions, replacements, modifications, partial deletions, and the like can be made to these embodiments without departing from the gist of the present disclosure or without departing from the gist of the present disclosure derived from the claims and equivalents thereof. Further, these embodiments can also be implemented in combination. For example, in the above-described embodiments, the order of operations and the order of processes are shown as examples, and are not limited to these. Furthermore, the same applies to a case where numerical values or mathematical expressions are used in the description of the above-described embodiments.

Claims

1. A water electrolysis system comprising: a water electrolysis device configured to electrolyze water; a gas-liquid separator configured to perform gas-liquid separation of a mixed fluid of hydrogen gas and water, the mixed fluid being led out from the water electrolysis device; a dehumidifier configured to dehumidify the hydrogen gas separated from the mixed fluid by the gas-liquid separator; a delivery path configured to deliver the hydrogen gas dehumidified by the

- dehumidifier; a humidifier configured to humidify the hydrogen gas delivered through the delivery path; and a compression device configured to compress the hydrogen gas humidified by the humidifier.
2. The water electrolysis system according to claim 1, wherein water obtained by dehumidifying the hydrogen gas using the dehumidifier is guided to any of the gas-liquid separator or the humidifier.
 3. The water electrolysis system according to claim 1, wherein the dehumidifier is a membrane dehumidifier.
 4. The water electrolysis system according to claim 1, wherein water obtained by any of the gas-liquid separator or the dehumidifier is guided to the humidifier.
 5. The water electrolysis system according to claim 4, wherein the gas-liquid separator includes a reservoir portion configured to store water, and the water electrolysis system further comprises a water supply unit configured to supply the water stored in the reservoir portion to the humidifier and the water electrolysis device.
 6. The water electrolysis system according to claim 5, wherein the water supply unit includes a heat exchanger configured to adjust a temperature of the water to be supplied to the humidifier and the water electrolysis device.
 7. The water electrolysis system according to claim 1, wherein the humidifier is a membrane humidifier.
 8. The water electrolysis system according to claim 1, wherein the delivery path is provided with a pump configured to send, to the compression device, the hydrogen gas guided from the dehumidifier.
 9. The water electrolysis system according to claim 1, further comprising an oxygen sensor configured to measure an oxygen concentration in the hydrogen gas flowing through the delivery path.
 10. The water electrolysis system according to claim 8, wherein the delivery path is provided with a branch path that branches off from a portion of the delivery path that is located on a downstream side of the pump, and water obtained by the dehumidifier is returned to the gas-liquid separator by the hydrogen gas flowing through the branch path.
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